

Lab 1: Deconstructing the C Compilation Pipeline

1. Objective

To analyze the transformation of C source code into an executable by isolating each phase of the GCC Toolchain: Preprocessing, Compilation, Assembly, and Linking.

2. Technical Theory

The transition from a .c file to a running program is a multi-stage pipeline. Each stage has a specific input and a specific output format.

- The Preprocessor (cpp): Handles "meta-programming." It replaces `#include` with the actual text of header files and swaps `#define` macros with their values.
 - The Compiler (cc1): Translates the preprocessed C code into Assembly Language. This is where syntax and semantic analysis happen.
 - The Assembler (as): Translates assembly mnemonics into Machine Code (binary instructions).
 - The Linker (ld): Connects your code to external libraries (like `printf` from the Standard C Library) to create a standalone executable.
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3. Lab Environment

- OS: Linux / Windows with MinGW
 - Compiler: GCC (GNU Compiler Collection)
 - Text Editor: VS Code
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4. Procedure & Manual Tracing

Step 1: Create the Source Code

Write the following code and save it as lab1.c.

```
#include <stdio.h>

int main() {
    int a, b, sum;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    sum = a + b;

    printf("Sum = %d\n", sum);

    return 0;
}
```

Step 2: Isolating the Phases

Run the following commands in your terminal and perform the required "Observations" for each file generated.

A. Preprocessing

Command: `gcc -E lab1.c -o lab1.i`

- Action: Open lab1.i in your editor.
- Observation: Note how the `#include <stdio.h>` has been replaced by thousands of lines of code.

B. Compilation (To Assembly)

Command: `gcc -S lab1.c -o lab1.s`

- Action: Open lab1.s.
- Observation: This is Assembly Language.

C. Assembly (To Object Code)

Command: `gcc -c lab1.c -o lab1.o`

- Action: Try to open lab1.o in a standard text editor.

- Observation: It will look like gibberish because it is binary (Machine Code).
- Advanced View: Run `nm lab1.o` in the terminal. This shows the "Symbol Table." You will see `main` and a reference to `printf`.

D. Linking

Command: `gcc lab1.o -o lab1.exe` (or just `lab1` on Linux)

- Action: Run the program with `./lab1`.

5. Write a program to calculate area of circle by defining PI as macro constant.